

OSAKA, Japan

WMO: 477720

Lat: 34.6822N

Lon: 135.5192E

Elev: 272

StdP: 14.55

Time Zone: 9.00 (E09)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	33.2	34.8	16.3	12.6	40.0	19.0	14.4	41.3	22.8	43.0	20.3	42.1	6.5	280	0.394

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	12.6	94.2	76.9	92.4	76.5	90.6	76.1	79.1	88.8	78.3	87.8	77.7	86.9	8.6	250

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
76.6	139.9	83.9	75.7	135.8	83.7	74.9	132.3	83.6	42.8	88.8	42.1	88.0	41.4	87.1	82.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 18.4	15.7	13.5	30.0	98.0	2.1	1.7	28.5	99.2	27.3	100.2	26.1	101.2	24.6	102.4		
(5)			25.8	80.6	2.4	0.7	24.1	81.1	22.6	81.5	21.3	82.0	19.5	82.5		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	63.3	43.0	44.0	49.8	59.6	68.7	75.1	82.7	85.2	78.0	67.8	57.0	47.6
	DBStd	15.25	3.70	4.73	5.41	5.80	4.34	3.99	4.14	3.40	5.01	4.95	5.25	4.68
	HDD50	583	220	179	72	4	0	0	0	0	0	0	4	104
	HDD65	2751	683	589	472	178	17	0	0	0	0	27	244	541
	CDD50	5440	2	10	65	291	579	752	1014	1092	840	552	214	29
	CDD65	2130	0	0	0	15	130	302	549	627	390	113	4	0
	CDH74	20126	0	0	0	43	535	1862	6149	7796	3325	413	3	0
	CDH80	7885	0	0	0	2	63	430	2542	3715	1086	47	0	0
Wind	WSAvg	6.0	6.5	6.1	6.3	6.2	6.0	5.9	6.3	6.4	5.9	5.4	5.2	5.9
Precipitation	PrecAvg	52.5	1.8	2.4	4.0	4.7	5.6	7.6	6.3	4.1	6.4	4.6	2.8	1.8
	PrecMax	68.6	4.8	7.7	6.9	9.4	14.2	15.9	17.6	13.4	19.1	16.9	6.7	4.1
	PrecMin	29.4	0.0	0.4	0.6	1.0	1.0	2.3	1.2	0.3	1.2	0.9	0.2	0.1
	PrecStd	9.1	1.1	1.4	1.5	1.9	2.9	3.2	3.8	3.0	3.8	3.1	1.6	1.1
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	57.5	63.8	69.4	78.5	84.9	90.2	96.0	97.2	92.8	84.7	73.7	63.4
		MCWB	49.9	54.2	56.0	62.3	64.9	74.2	77.4	77.4	76.0	71.1	61.6	55.1
	2%	DB	53.8	58.5	65.5	74.7	81.7	86.7	93.4	95.0	90.4	80.8	70.4	59.8
		MCWB	46.4	49.8	54.0	59.7	64.1	72.1	76.9	77.1	74.8	68.6	60.4	52.4
	5%	DB	51.7	54.8	62.5	71.9	79.2	84.5	91.6	93.4	88.3	78.4	68.0	57.6
		MCWB	44.7	46.9	52.3	58.6	63.6	70.5	76.4	76.7	74.0	66.5	59.1	50.1
	10%	DB	49.7	52.1	59.5	69.3	77.0	82.4	89.7	91.7	86.0	76.0	65.7	55.5
		MCWB	42.8	45.1	50.8	57.5	62.7	69.7	75.9	76.6	72.8	65.0	57.4	48.3
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	52.9	56.4	60.3	67.9	72.0	77.1	79.8	80.4	78.5	74.3	65.8	58.5
		MCDB	56.1	61.2	65.3	73.2	77.2	86.1	90.4	90.1	87.3	80.5	68.6	61.2
	2%	WB	48.3	52.0	56.8	64.1	69.4	75.5	78.8	79.5	77.3	71.3	63.0	54.2
		MCDB	52.1	56.3	62.2	69.4	75.1	82.9	88.4	89.2	85.7	78.1	67.9	58.4
	5%	WB	45.7	48.6	54.4	61.8	67.2	74.3	78.0	78.8	76.3	69.0	60.9	51.3
		MCDB	50.6	53.0	60.6	68.8	73.9	81.0	87.3	88.5	84.5	75.8	66.1	56.5
	10%	WB	43.5	45.8	51.8	59.6	65.5	72.7	77.3	78.2	75.3	66.7	58.7	48.9
		MCDB	49.0	51.4	58.5	66.8	73.1	79.2	86.1	87.7	83.4	73.7	64.6	54.3
Mean Daily Temperature Range	5% DB	MDBR	10.6	11.7	13.3	14.7	14.5	12.4	11.6	12.6	12.4	12.6	12.2	11.1
		MCDBR	13.2	15.8	17.5	18.0	17.3	15.2	13.3	14.0	14.5	14.7	14.2	13.7
		MCWBR	9.0	10.4	11.2	9.3	7.5	6.0	4.4	4.7	5.3	7.0	8.5	8.9
		MCDBR	11.9	13.4	14.9	14.3	13.2	11.6	12.3	12.3	12.5	12.4	12.0	12.2
	5% WB	MCWBR	10.0	11.1	12.1	9.8	8.8	5.3	4.5	4.6	5.4	7.9	9.3	10.0
Clear-Sky Solar Irradiance	taub	0.407	0.449	0.512	0.544	0.549	0.569	0.571	0.538	0.493	0.452	0.423	0.402	
	taud	2.126	2.028	1.887	1.860	1.913	1.935	1.980	2.059	2.155	2.184	2.169	2.155	
	Ebn at Noon	243	246	241	240	241	236	234	240	245	243	236	236	
	Edh at Noon	39	48	59	64	62	60	57	52	45	40	37	36	
All-Sky Solar Radiation	RadAvg	705	908	1226	1501	1648	1488	1580	1617	1274	1028	777	640	
	RadStd	50	72	96	131	170	141	206	143	107	101	64	41	

Historical Trends

		Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
(46) Station Only	DBAvg	(d) N/A	(e) N/A	(f) N/A	(g) N/A	(h) N/A	(i) N/A	(j) N/A	(k) N/A	(l) N/A
(47) Regional (33 neighbors)		(d) N/A	(e) N/A	(f) N/A	(g) N/A	(h) N/A	(i) N/A	(j) N/A	(k) N/A	(l) N/A

Nomenclature: See separate page